

	LSVEX	FMAGX	GSCGX	BRK
Alpha	0.00%	−0.27%	−0.14%	0.22%
t-stat	0.01	−2.23	−1.33	0.57
<i>MKT</i> Loading	0.94	1.12	1.04	0.36
t-stat	36.9	38.6	42.2	3.77
<i>SMB</i> Loading	0.01	−0.07	−0.12	−0.15
t-stat	0.21	−1.44	−3.05	−0.97
<i>HML</i> Loading	0.51	−0.05	−0.17	0.34
t-stat	14.6	−1.36	−4.95	2.57
<i>UMD</i> loading	0.2	0.02	0.00	−0.06
t-stat	1.07	1.00	−0.17	−0.77

The only alpha that is statistically significant is Fidelity Magellan, which is −0.27% per month or −3.24% per year. Poor investors in Fidelity lose money, and their losses are statistically significant. Berkshire Hathaway's alpha estimate is positive but insignificant. Our analysis in section 3.1 had a significantly positive alpha but we started in 1990. Now, starting ten years later in 2001, we don't even obtain statistical significance for Buffett. Detecting statistical significance of outperformance is hard, even in samples of more than ten years.

Looking at the factor loadings, LSV seems to be a big value shop—with a large *HML* loading of 0.51 (with a massive *t*-statistic of 14.6). Berkshire Hathaway is still value, too, with an *HML* loading of 0.34. Fidelity is a levered play on the market, with a beta of 1.12. Since none of the *UMD* loadings are large or significant, none of these funds are momentum players.

Style analysis seeks to rectify two potential shortcomings of our analysis so far:

1. The Fama–French portfolios are not tradeable.¹³
2. The factor loadings may vary over time.

Style Analysis with No Shorting

Style analysis tries to replicate the fund by investing passively in low-cost index funds. The collection of index funds that replicate the fund is called the “*style weight*.”

To illustrate, let's take the following index ETFs (see chapter 16 for 40-Act funds):

¹³ GM Asset Management has implemented tradeable versions of the Fama–French portfolios. See Scott (2012) for further details and chapter 14 on factor investing. Cremers, Petajisto, and Zitzewitz (2012) argue that the nontradeability of the Fama–French indices leads to distortions in inferring alpha.